

What the supply-shock exercise adds to the paper

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One question, answered plainly: what did running 189 supply-shock cells (`supply_etas_s1/prog/unif`, all executed, all gates pass) buy the paper? Short answer: it turns a *defect-looking degeneracy* into a *measured identification result*, it prices the elasticity that the matrix cannot see, and it restores the BETA row as an informative part of the closure family. It does not change a single demand-side number.

1 The problem the matrix left on the table

The published 5x3 matrix is demand-only: a green-investment programme shifted final demand, nothing else. Under demand-only shocks the model's price block is pinned by technology alone (the zero-profit conditions are homogeneous in prices and the wage, contain no demand term; Lemma 1 of `paper/equivalence.tex`), so relative prices stay at one and the real wage sits at its anchor. Two consequences look like defects:

- **BETA reproduces ALPHA bit-for-bit.** The elastic-supply row returns exactly the full-employment row, because with the real wage pinned the supply curve returns baseline employment whatever its elasticity.
- **eta_s (the labour-supply elasticity) is unidentified.** Every value 0.5, 2, 5 gives the same solution to machine precision. Reviewer point **R2.1** (labour-supply elasticity as bridging parameter) is precisely this demand.

The supply-shock exercise is the experiment that resolves the demand: the paper always knew the only channel that moves the real wage is technology (DE-0011 documented that the numeraire is *not* the lever). The arm makes technology shocks a first-class scenario and measures the wage-employment response.

2 What was measured

Three shock shapes, on the same full-71-sector calibration and the same 40.3 bn programme ($G0 = 40,300$ EUR m at 2019 prices, ADR-0024), 63 cells each, F1/F2/F3 financing, identification matrix ALPHA + scalar BETA (`eta_s` = 0.5/1/2/5) + sectoral uniform BETA:

Design	Shock	Wage (F2, mid ladder)	max abs(p-1)	Implied <code>eta_s</code>
<code>supply_etas_s1</code>	+10/20/30 % productivity in sector 1	1.0049	0.196	exact (0.5000/1.0000/2.0000/5.0000)
<code>supply_etas_prog</code>	+5/10/20 % on the programme's own sectors	1.0046	0.0625	exact
<code>supply_etas_unif</code>	+1/2/3 % in every sector	1.0408	0.0446	exact

$\ln L / \ln w$ recovers the input elasticity to four decimals in every scalar BETA cell and is invariant to the shock magnitude (spread < 1e-6 for s1, < 1e-3 for prog/unif). The ALPHA control pins employment at exactly 1 with the wage moving. All three arms satisfy the model's own accounting: F2 = F3 financing neutrality and the closed external-account identity hold under the technology shock on all 189 cells (the identity had previously only ever been asserted at prices equal to one).

3 What it adds

1. The identification result — η is a real parameter, and the null was the experiment, not the model. The map $\eta \rightarrow (\text{wage, employment, real GDP})$ is strictly monotone and invertible under a supply shock, and the model recovers the input elasticity *exactly* from its own response. The demand-only degeneracy is now explained as a property of the demand-only *design* (Lemma 1), not a hidden failure of the closure. That is the direct, measured answer to R2.1.

2. The sensitivity of the published results to the elasticity. The band that the matrix could not resolve is now a quantitative statement: under one representative technology shock the employment effect spans +0.24 % ($\eta = 0.5$) to +2.48 % ($\eta = 5$) — a 2.2 percentage-point spread attributable to the labour-supply assumption alone, and real GDP spans a similar range. The paper can state how much of its headline rests on the assumed elasticity per financing column, instead of leaving the parameter as an undisclosed dial.

3. BETA becomes an informative row. Under a technology shock BETA no longer equals ALPHA: employment rises with η while full employment stays flat, and prices move. The closure family shows variety exactly where the theory says it should. The “BETA is a free rider / the model cannot distinguish closures” objection loses its force.

4. The distribution of a productivity dividend between wages and jobs. The uniform arm is the cleanest reading: a 2 % economy-wide productivity gain raises the real wage 4 % and, depending on the labour-supply regime, yields real GDP growth of about +8 % ($\eta = 1$) to +27 % ($\eta = 5$) as employment expands from +4 % to +22 %. In the paper’s two-paradigm framing: a full-employment labour market (ALPHA) takes the whole dividend in wages; an elastic supply shares it with jobs. The exercise quantifies the elasticity of that distribution — a distributional statement the demand-only matrix could not touch.

5. A green-productivity scenario. The programme-own-sector arm (*prog*) asks: what if the decarbonisation programme itself raises productivity in construction and materials (learning, prefabrication) by up to 20 %? The answer includes a supply-side offset to the cost side — prices respond ($\max \text{abs}(p-1) = 0.0625$) and real GDP rises above the demand-only picture. That ties the exercise to the paper’s motivation (investment in foundational infrastructures) and to the climate-productivity idea recorded as IDEA-0001.

6. Architectural robustness, demonstrated, not assumed. Two properties the paper already leans on — financing neutrality ($F2 = F3$) and the closed external account — are verified under a second family of shocks on 189 cells. The $F2/F3$ finding is not an artifact of the demand-only design; the accounting is not fragile to $A \neq 1$.

7. The recombination, realized. The matrix deliberately left out the (η, η) corner (immobile allocation with elastic supply) as a demand-only null result. Under supply shocks that corner is informative, and the arm separates the two margins empirically: the allocation friction moves composition, the supply elasticity moves the level — exactly as the note anticipated.

4 What it could add, if drafted into the manuscript

Concrete, ready-draftable paragraphs (decide which earn their place):

- **The R2.1 referee paragraph:** one page stating the identification result, the Lemma-1 explanation of the demand-only equality, and the measured recovery table — with the flow tables cited by run id.
- **The disclosure paragraph:** the η -sensitivity band, reported per financing column, with the sentence “the employment spread across the elasticity grid [X pp] is the part of the result that rests on an assumption the demand-only data cannot pin down”.
- **The distributional reading** (item 4 above) — the strongest narrative hook for the heterodox framing: labour-market institutions determine who gets the productivity dividend.
- **The green-productivity scenario** (item 5) as a supplementary scenario alongside IDEA-0001.

5 What it does not add (so the claims stay honest)

- It does **not** repair the demand-only matrix: prices stay at one there by Lemma 1, and demand-sensitive prices in *that* matrix remain the job of the labour door and the capacity channel — the supply arm is a different experiment, clearly labelled as such.
- It does **not** estimate a real-world η from data. It builds the identification machinery and measures the band; an empirical elasticity (from an observed wage-employment response, e.g. a construction-sector shock) enters the model as a scenario, as it should.

- The `unif` magnitudes are partly architecture: the constant-returns open structure lets elastic supply absorb productivity gains aggressively. Report them as levels, do not over-read the multiplier.
- None of the demand-side matrix numbers changed anywhere (the generation re-ran bit-identical).

6 Where the numbers live

Flow tables `paper/tables/supply_etas_{s1,prog,unif}_flows.md` (generated from the manifests, no hand-typed numbers); runs `runs/supply_etas_s1-*`, `runs/supply_etas_prog-*`, `runs/supply_etas_unif-*`; the workplan and signatures in `docs/ETAs.md` v5; the assessment note in `docs/DOCS_ASSESSMENT.md` v8 (finding 2 upgrade); the schema decision in ADR-0023, the programme-size settlement in ADR-0024.

7 Revision Log

- **Version 1** (September 2026)